D4T4 DR1V3N RFI Response to CMS-2025-0050-0031

Revolutionizing the Health Technology Ecosystem with CMS

Submitted by: D4T4 DR1V3N

Contact: Melissa Williams, CEO

melissa.williams@theddinc.com

Executive Summary

D4T4 DR1V3N, a woman - and veteran-owned small business, commends CMS for advancing a secure, interoperable, and patient-centric health data ecosystem. Our federalized, software-only infrastructure is engineered with an offline and distributed first-framework accelerating the modernization of critical IT infrastructure with unparalleled resiliency & efficiency.

Powered by code626's (626) revolutionary software framework, D4T4 DR1V3N (DD) establishes a common distributed data format enabling 100% lossless data interoperability for all new instances of the Health Technology Ecosystem (HTE). With fully distributed, partitioned, and sharded data, the new HTE will provide centralized access to all health data, and resilient operations across mobile, rural, DDIL (Denied, Degraded, Intermittent, and Limited), and all 160 million Medicare patients due to DD's offline first architecture. STITCH™, our proprietary interoperability protocol, delivers unified, metadata-rich health records that transcend today's fragmented systems to empower all CMS stakeholders with seamless access to high-integrity data and unified digital capabilities leading to accelerated innovation and software development, informed health decisions, reduced burdens, & improved outcomes through new continuous and preventative care capabilities.

D4T4 DR1V3N provides the revolutionary HTE infrastructure, the building blocks, to view, manage, utilize, and share health data through any modern delivery that our country desperately needs. By embedding automation into data movement architecture, we also evolve from 2D to 4D health records ensuring provenance, traceability, transparency, and real-time reporting and auditing to combat waste, fraud, abuse, and inefficiencies in care. This 4D data transformation supports secure access across all Medicare contexts in alignment with national interoperability goals, TEFCA, FHIR, and Executive Order 14028 to scale CMS digital infrastructure. The new HTE will provide continuous access to quality health care for all Americans, while reducing the

cost and burdens, through unlocking the power of our data and revolutionizing healthcare decision making.

Our Vision: Revolutionizing the Health Technology Ecosystem with CMS

1. Technical Architecture and Core Capabilities

A. Lossless Data Movement with STITCH™

Core to our ecosystem's foundation, DD introduces entity-based data normalization that retains complete fidelity and provenance:

- Zero-loss data transfer across cloud, edge, and offline environments
- Native compatibility with FHIR, JSON, XML, SQL/NoSQL, and hierarchical file systems
- Embedded metadata with cryptographic versioning for origin, execution context, and server state

B. Distributed-First, Hybrid-Ready Deployment

DD is architected for:

- Active-active regional synchronization
- Reliable operations in DDIL (Denied, Degraded, Intermittent, Limited) environments
- Real-time replication with blockchain-based audit trail
- Forward and backward compatibility

C. Built-in Compliance and Zero Trust Security

Our new HTE integrates two AI-driven engines enhancing compliance automation:

- **ACT™** (Automated Classification Tool): Classifies PHI, CUI, HIPAA and SDOH data in <0.2s
- **ICE™** (Intelligent Compliance Engine): Enforces FedRAMP High, NIST 800-53/171, HIPAA, FISMA and Zero Trust policies preemptively. Supports EOs 14028 & 14158

D. Interoperability by Default

DD includes full API and routing support for:

- Blue Button 2.0, SMART on FHIR (standalone and EHR-launched)
- Bulk FHIR exports and (g)(10) APIs
- OAuth2/OpenID credentialing

- TEFCA QHIN routing with immutable audit trails

2. Aligned Use Cases for CMS Piloting

We recommend CMS prioritize pilots that show DD's full interoperability and compliance stack in action:

- Longitudinal Medicare Access Portals
- Digital Patient Identity and Interoperability Hub
- ACO/MSSP Data Aggregation Dashboards
- Federated Clinical Quality Reporting
- Clinical Research and Inventory Stack
- Provider Directory Interoperability
- TEFCA-Enabled Mobile Apps

3. Why DD Is Different

- Software-Only: No hardware nor managed services. Deployable via CLI or containers
- Audit-First: Auto-versioned, blockchain-backed snapshots for Medicare and regulatory audits
- Mission-Ready: Compliant with Eos- 14028, 14158, 13813- FISMA, and FedRAMP High.
- AI-Native: Structured, traceable data optimized for AI/ML and predictive analytics

Conclusion

D4T4 DR1V3N, powered by code626's revolutionary framework626 infrastructure, stands ready to support CMS in transforming the nation's health technology ecosystem. By delivering 100% lossless, metadata-rich, and fully interoperable data, our software-only, audit-first solution directly addresses the interoperability, compliance, and resiliency challenges outlined in this RFI. With built-in AI classification and compliance engines (ACT™ and ICE™), real-time distributed synchronization, and offline-first capabilities, we offer CMS a scalable and secure foundation aligned with FHIR, TEFCA, and Executive Order 14028. We welcome the opportunity to pilot this framework to help CMS lead a new era of patient-centric, value-based care, empowering Medicare beneficiaries with timely, accurate, and accessible health data—anywhere and at any time with evidence or data as our guide.

Responses to Key Sections of the RFI

PC-1: Health Management and Care Navigation App Recommendations

- **Longitudinal Patient Record**: A unified portal aggregating all Medicare records, claims, labs, visits, and treatments into a single, interactive timeline regardless of Provider or EHR. Helps patients & providers understand comprehensive health history, medication interactions, and care gaps across multiple providers.
- **Digital Patient Identity & Eligibility Verifier**: Uses biometric authentication, FHIR APIs, and OpenID-compliant credential to confirm identity, insurance coverage, and prior authorizations, even offline or in emergencies.
- **Medicare ACO Engagement Dashboard**: Visualizes cost, quality metrics, and patient-reported outcomes in real-time; allows beneficiaries to opt-in to preferred providers or care paths for chronic disease management.
- **TEFCA-Enabled Mobile Health Wallet**: A secure, offline-capable app for storing and sharing encrypted access tokens with Providers to securely transfer digital health passports, immunization records, and emergency instructions. Especially critical in DDIL scenarios.
- **AI-Powered Pharmacy & Drug Shortage Manager**: Provides up-to-date inventory and alerts about prescription availability, cheaper alternatives, or interaction risks- especially useful during disasters or supply chain disruptions.

Each of these apps, powered by framework626, would deliver a seamless, patient-centric experience aligned with CMS' goals to reduce fragmentation, enhance access, and improve health outcomes across diverse populations and care conditions.

PC-2: Data Access – One System, All Data

- No, most individuals- including caregivers - do not have easy access to their own or their loved ones' health information in a single, unified location. The current landscape is fragmented, requiring multiple logins across provider portals, inconsistent record formats, and redundant intake processes. This creates barriers to informed decision-making and coordinated care.
- The STITCH data protocol enables seamless access to all data types - demographics, claims, imaging metadata, prescriptions, and logs- in a lossless, versioned, and portable format. Patients, caregivers, and authorized providers can operate with the confidence that their data integrity is preserved and accessible, even in disconnected or degraded network environments.
- Contexts and workflows where utilizing a single, unified portal and longitudinal patient record include chronic disease management, family caregiving - managing records for elderly parents, children, or dependents with disabilities, emergency and disaster response, post-acute or transitional care, and benefits management/prior authorization.

PC-4: What's Missing from Current Apps

Today's digital health products often:

- Lack a universal patient profile with a comprehensive, longitudinal record across all providers, plans, and health systems
- Lack offline capabilities - disrupting continuity of care in rural, disaster-prone, or underserved areas
- Struggle with auditability and traceability of PHI - especially during transfers between vendors
- Few apps clearly communicate data provenance, access logs, or consent controls to the user
- No blockchain-style auditability **or** Zero Trust posture visible to end users

PC-5: Encourage patient and caregiver interest in digital health products

- Incentivize use through copay reductions, bonus coverage for chronic disease management if digital tools are routinely utilized
- Partner with community health centers, pharmacies, and AARP to educate beneficiaries on the value of digital health tools via trusted messengers.
- Publish performance metrics showing how apps improve medication adherence, reduce ER visits, or lower long-term costs
- Simplify access and identity management through a unified, government-backed digital ID system (e.g., Medicare Health Wallet) for patients and caregivers, with role-based permissions and proxy management

PC-8 / PC-12: Data Availability, Interoperability, APIs

- Readily available and valuable data includes:
  - Basic Claims Data (Part A/B/D): Billing codes, services, and coverage status via Blue Button 2.0
  - Patient Demographics
- High value data for patients and caregivers
  - Allergies, Medications, Vaccination records, lab test results, imaging exams from EHRs, public health systems, and portals constitute
  - Encounter / SOAP notes and operative reports often unstructured and thus excluded from FHIR APIs
  - Real-time appointments and scheduling availability
  - Price Transparency data: fragmented, inaccurate, and inconsistently presented.
- CMS data outside of Part A/B/D claims- such as encounter notes, clinical lab results, and real-time provider availability- should be accessible via FHIR bulk endpoints and FHIR (g)(10) gateway queries. code626 is ready to parse and export this data without middleware or format reconfiguration
- TEFCA-aligned APIs + STITCH enable bi-directional routing of beneficiary data across national provider directories and patient-selected applications

PR-1: Provider Adoption of Digital Health Products

- Solutions like **DD's**, which are offline-capable, interoperable by default, and designed for DDIL environments, directly address rural provider needs while supporting CMS-approved digital health adoption. The ability to work offline represents the most challenge obstacle and seamlessly addressing this shortcoming, providing a reliable solution to improve data integrity and outcomes would drive provider adoption
- Providers must articulate the impact and benefits of an offline capable solution delivering a comprehensive patient record to improving outcomes and transitioning to preventative care models

PR-2: Obstacles to Development, Deployment, and Utilization

- Data Fragmentation Across Systems
  - Problem: Lack of consistent access to complete, structured clinical and claims data hinders accurate quality reporting and documentation
  - Mitigation: Enforce FHIR API adoption across all certified EHRs and payers; mandate participation in TEFCA QHINs
- Poor Interoperability and Workflow Integration
  - Problem: Tools are often built as bolt-ons with little to no alignment with provider EHRs, increasing administrative friction
  - Mitigation**:** Require SMART on FHIR app certification and integration with provider workflows (e.g., embedded buttons or hooks in EHR UI)
- High Documentation and Administrative Burdens
  - Problem**:** Physicians spend significant time on manual charting and duplicative entries across systems
  - Mitigation**:** Promote adoption of AI-driven documentation tools with voice input, ambient listening and auto-classification (e.g., D4T4 DR1V3N's ACT™ for <0.2s classification). Our system replaces legacy EHR integrations with containerized apps that deliver:
    - Automated compliance (e.g., RMF, DD254, ATO)
    - Simplified quality metrics extraction via STITCH
    - SBOMs and security audits generated automatically during build, reducing dev & compliance fatigue
- Compliance and Legal Uncertainty
  - Problem: Concerns over HIPAA, billing audits, or ATOs delay innovation and create overhead
  - Mitigation**:** Expand CMS and ONC-approved "pre-certified" digital product registries with compliance attestation support like ICE™
- Rural and DDIL Barriers
  - Problem**:** Limited bandwidth, on-site IT support, and outdated hardware in rural areas stall deployment
  - Mitigation**:** Fund offline-first, container-based tools with active-active sync (e.g., STITCH™) that reduce need for cloud only reliance

PR-3: Importance of Full EHR Data Accessibility

- Current challenges in accessing different data formats:
  - Fragmentation of data types: Scanned documents, faxes, free-text notes, and structured fields often reside in siloed modules within EHR systems or are stored in proprietary formats, preventing unified access
  - Legacy infrastructure: Many health IT systems lack support for modern data standards (e.g., FHIR), limiting seamless integration and extraction of diverse data formats
  - Inconsistent metadata and provenance: Without unified tracking, it's difficult to reconcile the origin, version, and context of data, especially from external or historical sources
- Impact on patient care quality:
  - Incomplete clinical picture: Clinicians may miss critical insights (e.g., allergies noted in scanned intake forms or historical trends in free-text progress notes), leading to diagnostic errors or suboptimal treatment.
  - Delays in care coordination: Difficulty accessing full data hinders timely handoffs and comprehensive treatment planning across providers, disproportionately impacting rural and underserved communities
- Technical barriers to full data accessibility:
  - Incompatible data representations: Data stored in non-interoperable formats (PDFs, images, proprietary schemas) require advanced parsing or manual entry, increasing errors and costs
  - Lack of standardization: Absence of universally enforced schemas across systems makes real-time data exchange complex
  - Scalability constraints: Traditional architectures struggle to support real-time access and traceability at scale across distributed care networks
- Cost or privacy implications of making all data formats interoperable:
  - Cost: Converting legacy data formats into interoperable structures involves upfront investment in data transformation tools and staff training. However, long-term benefits include reduced duplication, streamlined audits, and improved billing accuracy
  - Privacy: Ensuring HIPAA and FISMA compliance when exposing historical, unstructured data (e.g., scanned notes) necessitates robust access controls, encryption, and traceability mechanisms like those in code626's framework626 platform
- Priority level compared to other interoperability needs:
  - High Priority: True interoperability mandates not only exchange of structured clinical fields but all patient-relevant data**,** including unstructured and scanned records. In both urban and rural settings, especially during disconnected or degraded environments, comprehensive data access is foundational for care continuity
- **Recommendation**: Adoption of a data framework like 626's that ensure lossless, traceable, and format-agnostic data movement will play a pivotal role in meeting CMS' objectives for a unified digital health ecosystem. These tools bridge the

technical and operational gaps that currently fragment patient data across care landscapes

PR-4: Policy and Regulatory Enhancements

- Mandate API-based Administrative Access:
  - Extend existing FHIR API requirements to include standardized endpoints for administrative workflows such as:
    - Patient scheduling and provider availability
    - Intake form templates and pre-population fields
    - Appointment status updates and cancellations
- Incentivize Compliance Through CMS Rulemaking:
  - Expand the ONC Certification Program to certify APIs that expose administrative data and workflows, not just clinical data
  - Include administrative interoperability metrics in CMS quality reporting programs and value-based payment models
- Enforce Provider Portal Interoperability:
  - Require that provider-facing systems (EHRs, practice management systems) support bi-directional communication with authorized third-party apps using patient-initiated data access
- Enhance HL7 FHIR Resources for Scheduling and Admin Tasks:
  - Ensure widespread support for FHIR Appointment, Schedule, Slot, and Questionnaire resources
  - Standardize mappings for intake forms using Questionnaire Response and enable pre-population based on known demographics and clinical history
- Develop a Federated Provider Directory Standard:
  - Standardize APIs for accessing provider contact details, accepted insurance, subspecialties, and facility affiliations to power third-party app functionality
- Identity and Consent Management Standards:
  - Implement consistent OAuth2-based patient authentication and consent tracking frameworks that allow secure delegation to third-party tools while maintaining HIPAA compliance
- Encourage Integration of TEFCA with Administrative APIs:
  - Extend the Trusted Exchange Framework and Common Agreement (TEFCA) participation criteria to cover scheduling, intake, and communication endpoints
- Support Bulk Data and Patient-Authorized API Access:
  - Allow patients to authorize long-lived access tokens for apps to sync scheduling changes or form updates, minimizing repeated logins and enhancing user experience
- By aligning EHR vendors, provider organizations, and third-party developers under these enhanced policies and standards, the healthcare system can support a modern, consumer-facing experience akin to what patients expect in other industries like banking or travel.

PR-5 / PR-7: FHIR APIs for Providers and Workflow Integration

- We natively support:
  - Patient Access API
  - Provider Directory API
  - Provider Access API
  - Bulk FHIR with Group ID-based filtering
  - SMART on FHIR (full authorization workflow)
- STITCH Data Engine enables all APIs to function in lossless, hybrid cloud deployments including disconnected environments, uniquely enhancing resilience and compliance
- All interactions are logged with traceable metadata, allowing fine-grained audit, rollback, and validation compliant with HIPAA, FISMA, and FedRAMP standards
- Group-based filtering and AI pre-fetch capabilities
- **Recommendation:** CMS pilot integration of ICE™ as an enforcement wrapper to validate API exposure, ensuring all CEHRT- certified systems provide real patient access, not token endpoints

TD-3: Digital Identity Implementation

- Challenges
  - User burden and access barriers: Patients with low digital literacy or limited access to technology may struggle to obtain and use IAL2/AAL2 credentials
  - Integration complexity: EHRs, health apps, and provider portals must adapt to federated identity workflows, requiring engineering and compliance resources
  - Trust alignment: Differing levels of assurance (LOA) across credential providers may hinder cross-system interoperability
- Benefits
  - Enhanced security and fraud prevention: Strong digital identity reduces identity theft, unauthorized access, and medical record mismatches
  - Streamlined user experience: Enables single sign-on (SSO) across systems, including government and third-party health services
  - Trust in data exchange: Verifiable identities improve confidence in cross-network patient matching and record sharing
- Impact of Requiring IAL2/AAL2 Credentials
  - Significantly improves resistance to phishing, credential stuffing, and session hijacking
  - Supports Zero Trust principles and compliance with EO 14028 and NIST SP 800-63-3 guidelines
  - Enhances record linking accuracy across payers, providers, and apps by tying data to verified identities
  - May delay adoption among smaller providers or vulnerable populations without support mechanisms

- Impact of Mandatory Use of OpenID Connect (OIDC)
    - Standardizes identity workflows: Ensures all third-party apps and portals follow a common, secure, interoperable protocol
    - Facilitates consent and audit: OIDC scopes and tokens improve control over data sharing, supporting TEFCA and HIPAA compliance
    - Accelerates innovation: Reduces vendor-specific barriers, allowing digital health startups and public tools to integrate easily
- **Recommendation**: CMS and ONC should encourage adoption of OIDC-compliant, IAL2/AAL2 digital identities with incentives and technical assistance, while maintaining fallback options to avoid exclusion of underserved populations. Digital identity (ID.me, Login.gov) is already integrated into the code626 FHIR auth gateway. We support IAL2/AAL2 credentialing out of the box

TD-4: Open, Standards-based APIs

- CMS can better encourage use of open, standards-based, publicly available APIs by:
    - Mandating FHIR and SMART on FHIR compliance through ONC certification and CEHRT requirements
    - Tying incentive payments and reimbursement models to the use of publicly documented APIs
    - Publishing public interoperability scorecards rating vendors and providers on API openness and performance
    - Requiring open API support in federally funded contracts and TEFCA participation
    - Providing SDKs, test sandboxes, and technical assistance to accelerate third-party integration and adoption
- **Recommendation:**
    - ONC prioritize open API certification over front-end design
    - Mandate CEHRT APIs include audit logs, identity access verification, and bulk access triggers

TD-5: Nationwide Provider Directory

- DD fully supports API-first implementation and will align with FHIR, SMART on FHIR, and Bulk FHIR endpoints upon CMS request
- We advocate for:
    - A nationwide FHIR directory with NPI integration, maintained by a CMS-recognized public/private partnership
    - Enforcing open API adoption over proprietary lock-in to drive innovation and reduce EHR vendor gatekeeping

TD-11: Exporting EHI

- All patient and provider data in our system is instantly exportable in both human-readable and structured formats. Our approach exceeds existing requirements under §170.315(b)(10) by capturing:

    - Data source/version
    - Time of capture and transfer
    - Server and app context
    - Full relational and audit context

TD-19: Price Transparency

- DD supports enhanced transparency workflows, including:
    - Drug inventory + shortage prediction
    - Real-time analytics for FEMA/HHS resource utilization
    - Identity-linked cost models for patients navigating complex care
- **Recommendation**: CMS define minimum API standards for real-time cost data delivery per provider and procedure

VB-3: Enabling Value-Based Care with Smart Infrastructure

- We support value-based care by offering:
    - Real-time classification of sensitive data (**ACT/ICE** apps)
    - Predictive risk stratification
    - SDOH integration
    - Automated auditability and reporting without extra FTEs
    - Support for predictive analytics and trial routing via AI-ready, lossless data pipelines
    - Real-time performance metrics for ACOs and MSSP orgs
    - Obstacle Removed: Disjointed, manually reconciled quality reporting
    - Opportunity Created: AI-enhanced care coordination and fraud detection across CMS programs
    - Security, compliance, and consent management
    - Zero Trust architecture with audit logs**:** Meet HIPAA, FISMA, and 21st Century Cures mandates
    - Digital identity (ID.me, Login.gov + IAL2/AAL2 credentialing)

Contextual Imperative: Addressing Medicare and Medicaid Waste with D4T4 DR1V3N

Recent federal investigations have revealed deep-rooted inefficiencies in CMS programs. According to CMS Administrator Dr. Mehmet Oz, U.S. taxpayers lose an estimated $14 billion annually to Medicaid fraud, largely driven by dual enrollment errors and identity mismanagement. Simultaneously, Medicare Advantage audits highlight overbilling risks projected to exceed $40 billion per year.

These failures persist not from lack of regulation, but from gaps in traceability, data integrity, and real-time oversight. To address these systemic weaknesses, CMS needs platforms that embed integrity at the data layer.

DD delivers this foundation:

- **STITCH™** Immutable Data Lineage: Every transaction is cryptographically versioned and anchored with origin metadata creating a tamper-evident, fully auditable trail across Medicaid and Medicare systems.

- **ACT™** and **ICE™** Real-Time Compliance: The Automated Classification Tool and Intelligent Compliance Engine flag and restrict improper billing, unauthorized access, and HIPAA/NIST/TEFCA violations.

- **Digital Identity Integration**: Supports IAL2/AAL2 credentialing (Login.gov, ID.me) for secure, role-based access eliminating ambiguities in dual enrollment and reducing identity fraud.

As CMS strengthens enforcement under Executive Orders 14028, 14158, and 13813, DD offers a scalable, turnkey solution to reduce fraud, enhance data trust, and world class infrastructure for the decade ahead.

Contextual Imperative: Aligning DD with CMS's Vision Under Dr. Mehmet Oz

In his inaugural address as the 17th CMS Administrator, Dr. Mehmet Oz outlined a comprehensive vision to transform the agency's approach to healthcare delivery and "Make America Healthy Again" through four strategic priorities:

- Empowering Patients: Providing personalized solutions to help individuals manage their health and navigate the complex healthcare system
- Equipping Providers: Enhancing access to patient information and reducing administrative burdens to allow healthcare providers to focus on delivering quality care
- Eliminating Fraud and Waste: Implementing measures to identify and eradicate fraudulent activities that deplete resources meant for vulnerable populations
- Promoting Preventive Care: Shifting the healthcare paradigm from reactive treatment to proactive wellness and chronic disease management

DD directly supports these goals:

- Patient Empowerment: DD's user-centric design offers intuitive interfaces and tools that enable patients to access and manage their health data securely
- Provider Support: Through real-time data synchronization and streamlined AI-enabled workflows, DD reduces administrative overhead, allowing providers to concentrate on patient care
- Fraud Detection: The platform's AI and advanced analytics and monitoring systems proactively detect anomalies and potential fraudulent activities, safeguarding resources
- Preventive Health Focus: By facilitating secure, continuous, and comprehensive data analysis, DD supports initiatives aimed at early detection and management of chronic conditions

Integrating DD and code626 into CMS's infrastructure would accelerate Dr. Oz's vision delivering smarter, safer, more efficient healthcare while improving outcomes for 160M+ Americans.

Product Maturity & Strategic Alignment

DD offers a proven, federal-ready, deployable architecture built to accelerate software modernization across complex distributed federal environments. Designed with a distributed-first, offline-capable approach, we enable lossless data movement, security, and regulatory compliance across ALL CMS systems.

Key differentiators:

- Plug-and-Play Deployment: Installs with a single script, integrating with existing infrastructure without requiring major refactoring

- STITCH Protocol: The cornerstone of our framework, guarantees 100% data integrity, traceability, and forward/backward compatibility across platforms and clouds

- Modular Development: Reusable components (RETL, RCache, RAnalytics) speed compliant app development without reinventing core backend services

- Security & Compliance: Immutable audit logs, UUIDS for all data objects, and built-in, auto-enforced compliance rules support Zero Trust, EO 14028, and FISMA compliance

D4T4 DR1V3N is not middleware or legacy monolith— it's the next generation, distributed and offline-first infrastructure, purpose-built for CMS and Medicare's evolving regulatory and data exchange landscape. Let's Make America Healthy Again!